Subpixel Anchor Graph for Multimodal Clustering of Hyperspectral and LiDAR Data

Yang Liu, Shujun Liu, *Member, IEEE*, and Luxi Xiao

***Abstract*—Traditional anchor graph clustering (AGC) methods usually perform suboptimally when dealing with subspace similarity caused by spectral mixing and typically lack physical interpretability during anchor selection, relying on complex postprocessing steps. To address this issue, we introduce hyperspectral (HS) unmixing (HU) into the AGC framework, showing their inherent equivalence. Specifically, we proposed the subpixel AGC (SAGC) method, which explicitly models subpixel information from mixed spectra to endmembers for clustering. It allows seamless migration of HU methods to AGC tasks for determining the number of anchors and guiding anchor selection. To enhance anchor diversity while preserving subpixel information, we design the maximum anchor diversity selection strategy (MADSS). Finally, we apply this framework to HS–light detection and ranging (LiDAR) AGC, providing an implicit spatial regularization method based on unified abundances and an efficient solving strategy. To improve robustness in complex scenarios, we extend the method to its deep counterpart [deep SAGC (DSAGC)], effectively modeling spatial information and nonlinear features. The experimental results show that SAGC achieves performance comparable to state-of-the-art (SOTA) methods on three HS–LiDAR datasets, while DSAGC delivers significant improvements. Code is available at: https://github.com/Liujehong/SAGC**

***Index Terms*—Anchor graph learning, deep learning, hyperspectral (HS) unmixing (HU), multimodal remote sensing (RS).**

I. Introduction

With the advancement of Earth observation technologies, multisource remote sensing (RS) data have become increasingly abundant. As a result, leveraging the complementary information and heterogeneity among these multimodal datasets for geospatial analysis has emerged as a promising research direction [1], [2], [3]. For example, hyperspectral (HS) data provide rich material and texture information of surface objects, while light detection and ranging (LiDAR) offers elevation data [4]. Integrating data from different modalities enables unified analysis that can benefit object interpretation in regions of interest. Over the past decades, land-cover classification using multimodal data has become a central task in the RS field. In supervised multimodal classification, Dalponte et al. [5], Wei et al. [6], [7], and Hong et al. [8] have conducted successful studies by fusing HS and LiDAR data. Flex-MCFNet [9] achieves multimodal RS classification through a flexible data augmentation strategy.

Despite their success in RS multimodal data applications, these methods typically rely on manually annotated priors and large amounts of training samples, which are both time-consuming and labor-intensive. As a result, unsupervised clustering methods for multimodal data have attracted increasing attention in recent years. Such methods aim to partition samples based solely on the intrinsic characteristics of multisource data, thereby avoiding the dependence on labeled data. Due to the theoretical soundness and robustness of subspace clustering (SC) for high-dimensional data [10], SC has been successfully applied to HS data, as demonstrated in the works of Lei et al. [11], Chen et al. [12], and Cai et al. [13]. Subsequently, more graph-based extensions have been developed for HS SC to further improve clustering accuracy [14], [15]. Nevertheless, conventional SC methods fail to meet the demands of multimodal RS data clustering. Benefiting from the growing attention of the machine learning community to multimodal clustering, a family of multiview clustering methods has been proposed for joint clustering of general image data from different perspectives [16], [17], [18]. These approaches aim to exploit interview correlations to perform clustering without relying on additional annotations, among which the subspace-based approaches are referred to as multiview SC (MVSC) [19]. By treating HS and LiDAR data as different views, existing MVSC methods can be applied to heterogeneous RS data to explore a consistent subspace for clustering [20], [21].

However, due to the substantial differences between multimodal RS data and typical multiview image data, directly applying MVSC to HS–LiDAR remains highly challenging [22]. First, existing MVSC methods often overlook the prevalent spectral mixing phenomenon in HS images (HSIs). Due to limitations in observational conditions and scattering and diffraction effects, the spectrum of a single pixel is typically a mixture of multiple material components [23], [24]. For example, the spectrum of a tree pixel may simultaneously contain mixed spectra from water, soil, and the tree itself. This mixing effect not only exacerbates the similarity of subspaces between different land-cover categories but also interferes with

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Yang Liu is with the College of Computer Science and Cyber Security, Chengdu University of Technology, Chengdu 610059, China (e-mail: liujehong@outlook.com).

Shujun Liu is with the School of Computer Engineering, Chengdu Technological University, Chengdu 611730, China, and also with the Key Laboratory of Earth Exploration and Information Techniques, Ministry of Education, Chengdu University of Technology, Chengdu 610059, China (e-mail: suldier@outlook.com).

Luxi Xiao is with the College of Geophysics, Chengdu University of Technology, Chengdu 610059, China (e-mail: xiaoluxi3978@163.com).

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the construction of cross-modal consensus subspaces, leading to suboptimal pixel-level clustering performance [25], [26]. Second, MVSC requires constructing a self-similarity graph across all samples, which has a high computational complexity of $\mathcal{O}(N^2)$ or $\mathcal{O}(N^3)$, making it difficult to adapt to large-scale RS heterogeneous data scenarios [19]. Although some anchor-graph-based clustering methods significantly reduce complexity by replacing the full graph with an anchor graph [27], [28], the anchor selection lacks physical interpretability and fails to effectively address the spectral mixing problem. Meanwhile, such methods still rely on complex postprocessing and feature vector decomposition during the final clustering stage, which results in a lack of flexibility. On the other hand, although some HS unmixing (HU) algorithms have been successfully applied in recent years, they can decompose mixed pixels into a set of pure spectra (endmembers) and their corresponding proportions (abundances) using physical model assumptions such as the linear mixing model (LMM) [26], [29], [30], [31], [32]. However, HU is typically regarded as an independent task separate from clustering. In HS–LiDAR clustering, the spectral mixing problem is inherently present and cannot be ignored. Therefore, there is an urgent need to develop a clustering method for HS–LiDAR that can simultaneously model spectral mixing while maintaining efficiency.

To address the above challenges, we propose a novel subpixel anchor clustering algorithm. To the best of authors' knowledge, this is the first work to achieve HS–LiDAR data clustering at the subpixel scale. The core contribution of this article lies in the fact that, while spectral mixing at the pixel scale can lead to suboptimal clustering results, at the subpixel scale, this challenge is transformed into subpixel prior knowledge that promotes clustering. For example, similar land-cover types often exhibit similar spectral mixing patterns. (As shown in Fig. 1, the subpixel mixed spectral proportion vectors corresponding to pixels of the same class exhibit the same clustering characteristics.) Based on this motivation, we prove the equivalence of the HU problem at the subpixel level and the anchor graph clustering (AGC) problem, unifying them into an independent optimization problem. To select appropriate anchors at the subpixel scale, we propose the maximum anchor diversity selection strategy (MADSS) strategy, which employs greedy optimization to choose initial anchors, thereby balancing diversity and subpixel representativeness. For flexible and efficient label prediction, we introduce a clustering approach based on abundance angle distance (AAD), which directly performs clustering in the abundance space by leveraging the clustering characteristics of mixed patterns. Furthermore, to incorporate the spatial structural prior from LiDAR data, we innovatively extend the concept of abundance to LiDAR, referred to as unified multimodal abundance. We theoretically prove that learning unified multimodal abundance as an anchor graph is equivalent to implicit conditional sparse regularization in the abundance space. Notably, for solving the optimization objective, we provide an efficient alternating optimization strategy, which ensures that the algorithm complexity grows linearly with the number of samples, thus guaranteeing good scalability. Moreover, our method can be easily extended to deep learning frameworks, enabling the explicit modeling of nonlinear and spatial features in HS–LiDAR data. Overall, our main contributions are as follows.

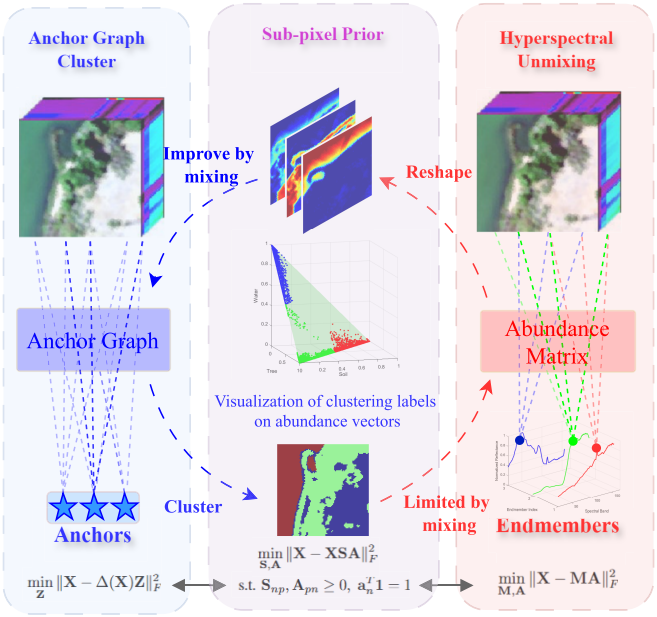

Fig. 1. Relationship between unmixing and anchor-based clustering is interdependent: while the clustering performance is constrained by the mixing, the subpixel information provided by unmixing can, in turn, enhance clustering results. The central figure illustrates our motivation. The visualization demonstrates that, by using different colors to represent the abundance vectors of different land-cover classes, the subpixel abundance vectors exhibit clear clustering patterns.

1) For the first time, we propose a subpixel scale AGC algorithm, a unified framework for HS–LiDAR data clustering. By formulating HU and AGC as a unified optimization problem, we innovatively transform spectral mixing, which would normally hinder clustering, into subpixel prior knowledge that benefits clustering.
2) Our framework offers excellent computational efficiency and scalability. We derive an efficient iterative optimization that guarantees linear growth in computational cost with the number of samples, enabling scalability to large datasets. Additionally, we propose clustering directly based on the mixed pattern characteristics, further ensuring efficiency.
3) Our method leverages the clustering properties of the abundance vectors to perform direct label prediction on the anchor graph, thereby eliminating the need for complex postprocessing steps while simultaneously enhancing the sparsity and separability of the learned anchor graph matrix.
4) We evaluate the proposed methods on multiple real-world datasets, demonstrating superior clustering performance and runtime efficiency. Extensive ablation and hyperparameter studies validate the motivation and robustness of our approach.

II. Preliminaries

A. Subspace Clustering

SC is an important unsupervised method that assumes high-dimensional samples within a cluster lie on a low-dimensional manifold or affine subspace, typically formulated as

$$\min_{\mathbf{C}} \|\mathbf{X} - \mathbf{XC}\|_F^2 + \alpha\Omega(\mathbf{C}), \quad \text{s.t. } C_{ii} = 0. \tag{1}$$

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The first term represents the subspace representation, in which each sample can be linearly expressed by other samples belonging to the same subspace, and samples within the same subspace are grouped into one cluster. Specifically, let the samples be denoted as $\mathbf{X} \in \mathbb{R}^{D\times N}$, where N is the number of samples and D is the data dimension, and let $\mathbf{C} \in \mathbb{R}^{N\times N}$ denote the coefficient matrix. After solving for the optimal $\mathbf{C}$, a series of graph postprocessing steps and spectral clustering [33] is performed to obtain the final clustering results. In HS imagery, different land-cover classes often exhibit discriminative low-dimensional subspaces, which SC methods can exploit to improve clustering accuracy. Existing works such as LRR [34] and SSC [10] employ low-rank or sparse regularization to better separate subspaces, while HS-specific approaches like NCSC [35] and GCOT [36] further leverage spectral correlation and spatial priors for improved results. IamCSC [37] achieves multimodal clustering for HS–LiDAR data by exploiting intuitive assimilated modalities. However, constructing the full affinity graph and spectral decomposition requires $\mathcal{O}(N^2)$ and $\mathcal{O}(N^3)$ time, respectively, limiting scalability.

To address this, anchor graph methods select a small set of representative samples (anchors) to approximate the association matrix, reducing complexity to $\mathcal{O}(MN)$

$$\min_{\mathbf{Z}} \|\mathbf{X} - \Delta(\mathbf{X})\mathbf{Z}\|_F^2 + \lambda\Omega(\mathbf{Z}). \tag{2}$$

Let $\mathbf{Z}$ represent the anchor graph, where $\mathbf{Z} \in \mathbb{R}^{M\times N}$, and $M \ll N$. $\Delta(\cdot)$ denotes theanchors' selection. Despite their efficiency, existing anchor-based methods face challenges in handling heterogeneous RS data. Spectral mixing causes subspaces of different land-cover types to become highly similar, hindering accurate modeling. Moreover, most methods are designed for single-modality data and cannot efficiently handle heterogeneous HS–LiDAR datasets. Therefore, it is essential to develop a robust AGC method capable of handling multimodal spectral mixed data.

B. Hyperspectral Unmixing

Spectral mixing is a ubiquitous phenomenon in HS data that limits its performance in downstream applications. This phenomenon arises from the limited spatial resolution during HS imaging, spectral overlaps among materials, and variations in observation conditions. By solving the physical model based on assumptions about the mixing mechanism, mixed pixels can be decomposed into several pure spectral signatures (endmembers) and their corresponding subpixel abundances. The HU problem using the most commonly adopted LMM can be represented as follows:

$$\begin{aligned}&\arg\min_{\mathbf{M},\mathbf{A}} \|\mathbf{X} - \mathbf{M}\mathbf{A}\|_F^2\\ &\text{s.t. } \mathbf{M}_{\text{lp}} \geq 0, \quad \mathbf{A}_{\text{pn}} \geq 0,\ \mathbf{a}_n^{\mathrm{T}}\mathbf{1} = 1.\end{aligned} \tag{3}$$

It describes that each pixel $\mathbf{X} \in \mathbb{R}^{L\times N}$ in HS data can be represented as a linear combination of pure spectra $\{\mathbf{M} = \{\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_P\} \in \mathbb{R}^{L\times P}\}$. The abundance matrix $\{\mathbf{A} = \{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N\} \in \mathbb{R}^{P\times N}\}$ indicates the proportion of each endmember present in a pixel. Each column $\mathbf{a}_n$ of $\mathbf{A}$ represents the abundance vector corresponding to a pixel and is referred to as the subpixel information. The constraints $\mathbf{M}_{\text{lp}} \geq 0, \mathbf{A}_{\text{pn}} \geq 0$, and $\mathbf{a}_n^{\mathrm{T}}\mathbf{1} = 1$ represent the physical constraints on abundances and endmembers.

TABLE I
SUMMARY OF GRAPH AND RELATIONSHIP ACROSS TASKS

Task	Graph	Relationship	Subpixel info
HU	$\mathbf{A}$: Abundance	Samples-Endmembers	✓
SC	$\mathbf{C}$: Full graph	Samples-Samples	–
AGC	$\mathbf{Z}$: Anchor graph	Samples-Anchors	–
Ours	$\mathbf{A}$: Anchor graph	Samples-Endmembers	✓

In the field of unsupervised clustering, many existing studies primarily use clustering as a constraint or guidance to improve unmixing performance [38], [39]. However, leveraging subpixel information derived from unmixing to enhance clustering has been largely underexplored. Furthermore, HU is typically treated as an independent preprocessing step, and this two-stage approach often leads to error accumulation and suboptimal performance. Therefore, this article is the first to approach the unmixing and clustering problem from a holistic perspective.

III. SUBPIXEL AGC

In this section, we unify AGC and unmixing within a single framework and, based on this, propose a method called subpixel AGC (SAGC). The pipeline of SAGC is illustrated in Fig. 2.

A. Unified Framework for Clustering and Unmixing

This article is the first to explore the intrinsic connection between HU and AGC. As illustrated in Table I, endmembers can be interpreted as anchors, while the abundance matrix represents the anchor-to-sample relationship, forming a subpixel anchor graph. We propose the following proposition.

Proposition 1: Equations (2) and (3) are equivalent.

Proof: We begin by considering the HU optimization problem, let $\mathbf{S}_0 \in \mathbb{R}^{N\times P}$ be an initialized endmember matrix, obtained from methods such as K-means or VCA [29]. We can describe endmember initialization in the following form:

$$\mathbf{M} = \mathbf{X}\mathbf{S}, \quad \text{s.t. } \mathbf{S}_{\text{np}} \geq 0 \tag{4}$$

where $\mathbf{S}_{\text{np}} \geq 0$ is to ensure the physical constraint of $\mathbf{M}$ is valid. Then, (3) can be rewritten as follows:

$$\min_{\mathbf{S},\mathbf{A}} \|\mathbf{X} - \mathbf{X}\mathbf{S}\mathbf{A}\|_F^2 \quad \text{s.t. } \mathbf{S}_{np}, \quad \mathbf{A}_{\text{pn}} \geq 0,\ \mathbf{a}_n^{\mathrm{T}}\mathbf{1} = \mathbf{1}. \tag{5}$$

By comparing (2) with (5), it can be observed that when endmembers are treated as anchors and the abundance map as the anchor graph, HU can be viewed as a special case of AGC with dynamic anchor updates. Furthermore, (4) can be considered a specific form of anchor selection $\Delta(\mathbf{X})$, which we refer to as subpixel anchor selection.

Accordingly, based on this proof, we propose a subpixel anchor graph learning method that unifies unmixing and clustering. We formulate our core optimization problem as

$$\min_{\mathbf{S},\mathbf{A}} \|\mathbf{X} - \mathbf{X}\mathbf{S}\mathbf{A}\|_F^2 + \alpha\|\mathbf{A}\|_F \quad \text{s.t. } \mathbf{S}_{\text{np}}, \quad \mathbf{A}_{\text{pn}} \geq 0,\ \mathbf{a}_n^{\mathrm{T}}\mathbf{1} = \mathbf{1} \tag{6}$$

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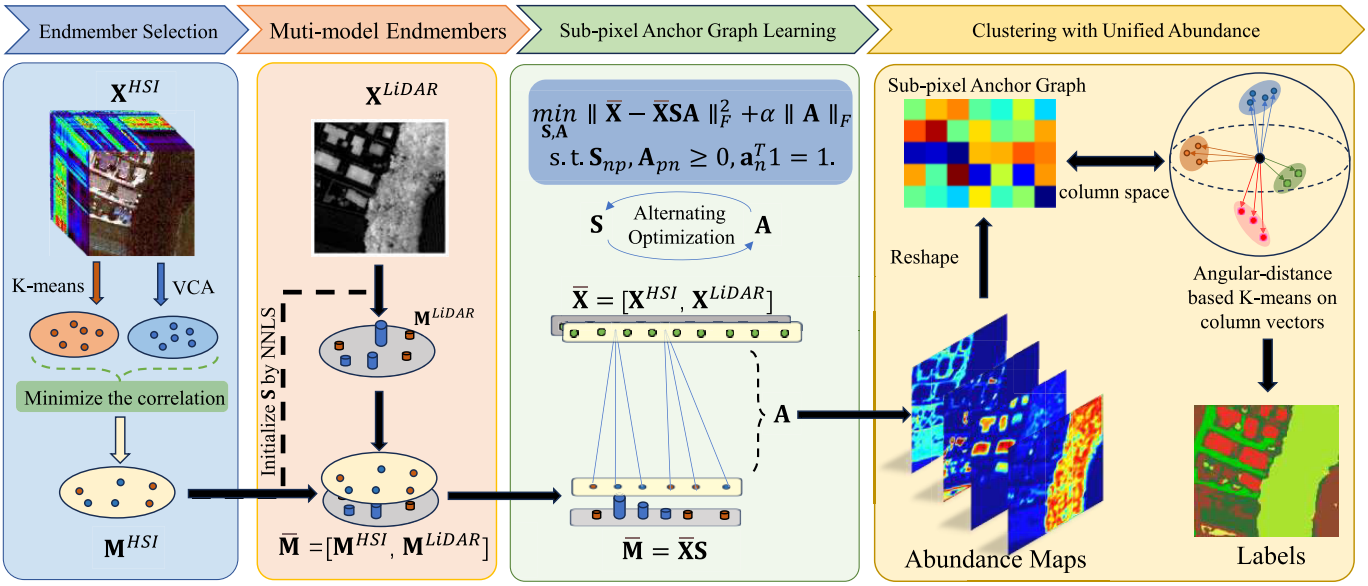

Fig. 2. Overall pipeline of our approach.

where $\mathbf{S} \in \mathbb{R}^{N\times P}$ and $\mathbf{A} \in \mathbb{R}^{P\times N}$. The second term is a regularization term that controls the sparsity of the abundance matrix, with α as its weight. This approach provides strong physical interpretability for adaptive anchor (endmember) updates. Once the abundance matrix is obtained, it can be directly used as an anchor graph for clustering. Moreover, it extends the theoretical boundary of existing AGC by enabling endmember extraction algorithms in unmixing to be generalized for anchor selection.

B. Anchor Selection

To realize SAGC, two core issues must be addressed: how to determine the number of anchors and how to initialize them. To avoid relying on manual expertise for anchor number selection, we leverage the theoretical connection between unmixing and AGC (Proposition 1) and employ HySime [40] to adaptively estimate the number of subpixel anchors. Furthermore, to balance pixel-level diversity and subpixel representativeness, we propose the MADSS, which selects the final anchors from a candidate endmember library constructed by K-means and VCA. This strategy effectively balances representativeness and diversity at the subpixel level, thereby ensuring that the initialized anchors are both reasonable and physically interpretable.

Specifically, let the VCA-derived endmembers be $\mathcal{V} = \{v_1, v_2, \ldots, v_P\}$ and the K-means centroids be $\mathcal{K} = \{k_1, k_2, \ldots, k_P\}$. The combined candidate set is defined as: $\mathcal{A}_{\text{cand}} = \mathcal{V} \cup \mathcal{K} = \{a_1, a_2, \ldots, a_{2P}\}$. Given the redundancy in $\mathcal{A}_{\text{cand}}$, we formulate the MADSS problem as

$$\mathbf{M} = \arg\min_{\substack{\mathbf{M}\subseteq\mathcal{A}_{\text{cand}} \\ |\mathbf{M}|=P}} \sum_{i\neq j} \text{sim}\left(a_i, a_j\right), \quad a_i, a_j \in \mathcal{A}_{\text{cand}}. \tag{7}$$

This problem is solved greedily by iteratively selecting one anchor a_p that is least similar to the current anchor set $\mathcal{A}_f$

$$a_p = \arg\min_{a} \sum_{b\in\mathcal{A}_f} \text{sim}\,(a, b) \quad \text{s.t. } a \in \mathcal{A}_{\text{cand}} \setminus \mathcal{A}_f. \tag{8}$$

After P iterations, we obtain the final anchor set $\mathbf{M} = \mathcal{A}_f$.

C. Unified Multimodal Abundance

To integrate spectral mixing priors into HSI–LiDAR joint clustering, we propose a unified multimodal abundance representation that enforces the spectral and elevation modalities to be reconstructed with the same set of coefficients, which is equivalent to an implicit sparse regularization. Specifically, after selecting anchors from the information-rich HSI using the MADSS strategy, the LiDAR information is subsequently incorporated into them. As shown in (4), these anchors can be linearly represented by the original data, thereby preserving their physical interpretability. Therefore, given the HSI anchors $\mathbf{M}^{\text{HSI}}$, we solve for their coefficients $\mathbf{S}$ using nonnegative least squares (NNLS) on (4). Leveraging the spatial alignment of LiDAR elevation data with the HSI, we further estimate physically meaningful elevation values $\mathbf{M}^{\text{LiDAR}}$ for each anchor. The final multimodal anchor $\bar{\mathbf{M}}$ construction is given as follows:

$$\bar{\mathbf{X}} = \text{concat}\begin{pmatrix}\mathbf{X}^{\text{HSI}} \\ \mathbf{X}^{\text{LiDAR}}\end{pmatrix} \tag{9}$$

$$\bar{\mathbf{M}} = \begin{bmatrix}\mathbf{M}^{\text{HSI}} \\ \mathbf{M}^{\text{LiDAR}}\end{bmatrix} = \bar{\mathbf{X}}\mathbf{S}. \tag{10}$$

This representation requires the algorithm to identify endmembers whose spectral and elevation components simultaneously satisfy the reconstruction constraint. Ideally, if the endmember library exhibits sufficient diversity in the elevation dimension, the algorithm can select appropriate endmembers to meet this condition. However, in practice, when endmembers are directly constructed from the data, the elevation diversity is often limited. As a result, a low reconstruction error can only be achieved when all activated endmembers (i.e., those with nonzero abundances) have similar elevation values to the target pixel. We refer to this condition as implicit sparse regularization

$$I = \{i \mid \mathbf{A}_{in} \geq 0\} \quad \forall i, j \in I,\ \mathbf{m}_i^{\text{LiDAR}} \approx \mathbf{m}_j^{\text{LiDAR}} \approx \mathbf{X}_n^{\text{LiDAR}}.$$

Therefore, the anchor graph learning process tends to activate endmembers/anchors that are consistent with the elevation.

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Since only endmembers consistent with the pixel's elevation are activated, the resulting abundance vectors are highly sparse, which enhances their separability for clustering. Additionally, due to the strong spatial priors in LiDAR-derived elevation maps, neighboring pixels often share similar elevation and spectral properties. This naturally restricts the activated endmembers to spatially coherent subsets, leading to a locally smooth abundance map, thereby incorporating the LiDAR spatial structure information into the anchor graph learning process.

D. Optimization Strategy

In this section, we present the optimization details of (5). Since the problem is nonconvex when considering both the matrices $\mathbf{S}$ and $\mathbf{A}$, we adopt an iterative approach to solve it. As suggested in [41], we employ the Lagrange multipliers to update $\mathbf{S}$ and $\mathbf{A}$.

First, by combining (9) with (10), we rewrite (5) as a function to be solved

$$\begin{aligned} f(\mathbf{S},\mathbf{A}) &= \mathrm{Tr}\left(\left(\bar{\mathbf{X}}-\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\right)\left(\bar{\mathbf{X}}-\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\right)^{\mathrm{T}}\right)+\alpha\|\mathbf{A}\|_F \\ &= \mathrm{Tr}\left(\bar{\mathbf{X}}\bar{\mathbf{X}}^{\mathrm{T}}\right)-2\mathrm{Tr}\left(\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\bar{\mathbf{X}}^{\mathrm{T}}\right) \\ &\quad +\mathrm{Tr}\left(\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\mathbf{A}^{\mathrm{T}}\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\right)+\alpha\sqrt{\mathbf{Tr}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)}. \end{aligned} \tag{11}$$

Due to the physical constraints in (6), specifically $\mathbf{S}_{\mathrm{np}}, \mathbf{A}_{\mathrm{pn}} \geq 0$, we introduce Lagrange multiplier matrices $\Phi_{\mathrm{np}} \geq 0$ and $\Psi_{\mathrm{pn}} \geq 0$, whose dimensions are the same as $\mathbf{S}$ and $\mathbf{A}$, respectively. Furthermore, considering the abundance sum-to-one constraint, the objective function can be rewritten in the following form:

$$\begin{aligned} \Gamma = {} & \mathrm{Tr}\left(\bar{\mathbf{X}}\bar{\mathbf{X}}^{\mathrm{T}}\right)-2\mathrm{Tr}\left(\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\bar{\mathbf{X}}^{\mathrm{T}}\right)+\mathrm{Tr}\left(\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\mathbf{A}^{\mathrm{T}}\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\right) \\ & \times\alpha\sqrt{\mathbf{Tr}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)}+\mathrm{Tr}\left(\Psi\mathbf{A}^{\mathrm{T}}\right)+\mathrm{Tr}\left(\Phi\mathbf{S}^{\mathrm{T}}\right) \\ & +\lambda_1\left(\mathbf{a}_1^{\mathrm{T}}\mathbf{1}-\mathbf{1}\right)+\cdots+\lambda_N\left(\mathbf{a}_N^{\mathrm{T}}\mathbf{1}-\mathbf{1}\right). \end{aligned} \tag{12}$$

Equation (12) indicates that incorporating the sum-to-one constraint requires introducing N scalar Lagrange multipliers λ_n, which severely affects the solving efficiency when N is large. To alleviate this problem, we adopt a matrix augmentation method that ensures the sum-to-one constraint is satisfied in parallel when updating $\mathbf{A}$.

Specifically, by omitting the Lagrangian term corresponding to the sum-to-one constraint, we take the partial derivatives of both sides of (12) with respect to $\mathbf{S}$ and $\mathbf{A}$, and obtain

$$\begin{cases} \dfrac{\partial\Gamma}{\partial\mathbf{S}} = -2\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{A}^{\mathrm{T}}+2\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\mathbf{A}^{\mathrm{T}}+\Psi \\ \dfrac{\partial\Gamma}{\partial\mathbf{A}} = -2\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}+2\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{S}\mathbf{A}+\dfrac{\alpha\mathbf{A}}{\|\mathbf{A}\|_F}+\Phi. \end{cases} \tag{13}$$

Due to the presence of inequality constraints, the necessary conditions for optimality can be derived by applying the *Karush–Kuhn–Tucker* (*KKT*) conditions

$$\begin{cases} \Psi\odot\mathbf{S}=\mathbf{O} \\ \Phi\odot\mathbf{A}=\mathbf{O}. \end{cases} \tag{14}$$

By substituting the result of setting (13) to zero into (14), we obtain

$$\begin{cases} \left(-2\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{A}^{\mathrm{T}}+2\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)\odot\mathbf{S}=\mathbf{O} \\ \left(-2\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}+2\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{S}\mathbf{A}+\dfrac{\alpha\mathbf{A}}{\|\mathbf{A}\|_F}\right)\odot\mathbf{A}=\mathbf{O}. \end{cases} \tag{15}$$

After simplification, the update rules for $\mathbf{S}$ and $\mathbf{A}$ are derived as follows:

$$\mathbf{S}^{t+1} = \mathbf{S}^{t}\odot\frac{\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{A}^{\mathrm{T}}}{\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{S}\mathbf{A}\mathbf{A}^{\mathrm{T}}} \tag{16}$$

$$\mathbf{A}^{t+1} = \mathbf{A}^{t}\odot\frac{2\|\mathbf{A}\|_F\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}}{2\|\mathbf{A}\|_F\mathbf{S}^{\mathrm{T}}\bar{\mathbf{X}}^{\mathrm{T}}\bar{\mathbf{X}}\mathbf{S}\mathbf{A}+\alpha\mathbf{A}}. \tag{17}$$

Subsequently, to efficiently embed the sum-to-one constraint into the multiplicative update framework, we incorporate this constraint into the update rules by replacing $\mathbf{S}$ and $\mathbf{A}$ in (16) and (17) with the following augmented matrices:

$$\bar{\mathbf{X}}\leftarrow\begin{bmatrix}\bar{\mathbf{X}}\\ \lambda\mathbf{1}_N^{\mathrm{T}}\end{bmatrix}\quad \bar{\mathbf{X}}\mathbf{S}\leftarrow\begin{bmatrix}\bar{\mathbf{X}}\mathbf{S}\\ \lambda\mathbf{1}_P^{\mathrm{T}}\end{bmatrix}. \tag{18}$$

In the multiplicative update rules, the incorporation of specific row vectors into the augmented matrix naturally enforces the equality $\mathbf{1}_N^{\mathrm{T}}=\mathbf{1}_P^{\mathrm{T}}\mathbf{A}$, thereby achieving the column sum-to-one constraint on $\mathbf{A}$.

E. Postprocessing and Prediction

Ultimately, the unified abundance matrix $\mathbf{A}$ can be viewed as a subpixel anchor graph that captures the relationships between samples and endmembers. However, traditional SC methods require several postprocessing steps to obtain the final labels, such as constructing the Laplacian matrix and performing eigenvalue decomposition (EVD) or singular value decomposition (SVD) [42].

Thanks to the physical interpretability of our method, the anchor graph $\mathbf{A}$ has clear physical meaning, specifically representing subpixel abundances. Naturally, abundance vectors corresponding to the same class of objects should be as similar as possible. For instance, a grassland pixel may consist solely of a mixture of three endmembers: water, soil, and grass. Based on this observation, we adopt the *AAD* as the similarity metric, defined as

$$\mathrm{AAD}\left(\mathbf{a}_i,\mathbf{a}_j\right)=\arccos\left(\frac{\mathbf{a}_i^{\mathrm{T}}\hat{\mathbf{a}}_j}{\|\mathbf{a}_i\|_2\left\|\hat{\mathbf{a}}_j\right\|_2}\right),\quad \mathbf{a}_i,\mathbf{a}_j\in\mathbf{A}. \tag{19}$$

We propose to directly perform K-means clustering in the column space of the anchor graph $\mathbf{A}$ based on AAD, thereby avoiding spectral decomposition and subsequent graph postprocessing.

F. Complexity Analysis

The pseudocode of the proposed SAGC method is presented in Algorithm 1. During the initialization phase, the major computational cost arises from three components. First, the Hysime algorithm used for estimating the number of endmembers incurs a total complexity of $\mathcal{O}(L^2N+L^3)$. Second, the MADSS module—which includes VCA, K-means, and a greedy sampling strategy—adds further cost. Specifically, VCA requires $\mathcal{O}(L^2N+L^3+P^2N)$, and K-means has a complexity of $\mathcal{O}(T_1\mathrm{NPL})$. Note that the additional computational cost of the greedy selection is negligible compared to that of K-means and VCA. Considering that $P\ll L\ll N$, the overall initialization complexity can be approximated as $\mathcal{O}(T_1\mathrm{NPL})$.

In the subsequent subpixel anchor graph learning stage, the naive computation of (16) requires $\mathcal{O}(N^2(L+P))\approx\mathcal{O}(N^2)$. To improve efficiency, we rearrange the computation order of the

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Algorithm 1 SAGC for HS–LiDAR Data

Input:
$\mathbf{X}^{\mathbf{HSI}} \in \mathbb{R}^{L\times N}$, $\mathbf{X}^{\mathbf{LiDAR}} \in \mathbb{R}^{1\times N}$, anchors ratio K, the number of land-cover classes D and maximum iterations T.
Output: Cluster labels.

Initialization:
Estimate number of anchors $P \leftarrow K \times \text{HySime}(\mathbf{X}^{\text{HSI}})$;
Apply MADSS on HSI to select anchors $\mathbf{M} \in \mathbb{R}^{L\times P}$;
Initialize the anchor selection matrix $\mathbf{S}$ apply NNLS on Eq. (4);
Concatenate $\bar{\mathbf{X}} = concat\begin{pmatrix}\mathbf{X}^{\text{HSI}}\\ \mathbf{X}^{\text{LiDAR}}\end{pmatrix}$;
Initialize the multimodel anchors by Eq. (10).

Iterative Optimization:
while *not converged* ***or*** $t < T$ **do**
  Update $\mathbf{S}$ using Eq. (16);
  Update $\mathbf{A}$ using Eq. (17);
Apply K-means based on Eq. (19) in the column space of $\mathbf{A}$ to obtain the cluster labels.

matrix multiplications by computing $\mathbf{G} = \bar{\mathbf{X}}\mathbf{A}^{\text{T}}$ followed by $\bar{\mathbf{X}}^{\text{T}}\mathbf{G}$, thereby reducing the complexity to $\mathcal{O}(\text{LNP})$. A similar approach applies to (17), also yielding $\mathcal{O}(\text{LNP})$ per iteration. Letting T_2 denote the maximum number of iterations, the total iterative complexity is $\mathcal{O}(T_2\text{LNP})$. After clustering with K-means into D clusters, the overall complexity of SAGC becomes $\mathcal{O}((T_1 + T_2)\text{NPL} + T_3\text{NDP})$. Given that $N \gg L \gg P \approx D$, the complexity grows approximately linearly with the number of samples, which enables us to cluster all the data at once, thereby ensuring scalability on large-scale HS datasets.

IV. DSAGC

While the proposed SAGC framework is formulated in an optimization-based manner, one of its key advantages is generality. The same principles can be seamlessly extended to deep neural architectures, leading to our deep variant termed deep SAGC (DSAGC). This extension allows the framework to exploit nonlinear feature modeling and spatial priors inherent in neural networks, thereby further improving clustering performance [32], [43], [44]. As illustrated in Fig. 3, we construct the deep version DSAGC using a lightweight dual-autoencoder network as an example. This architecture consists of two fully convolutional encoders, which can efficiently extract nonlinear spatial features from both modalities without introducing computationally intensive optimization.

The forward propagation of the two encoders is defined as

$$\mathbf{F}^{\text{HSI}} = \phi\left(\mathbf{X}^{\text{HSI}}\right) \tag{20}$$

$$\mathbf{F}^{\text{LiDAR}} = \psi\left(\mathbf{X}^{\text{LiDAR}}\right) \tag{21}$$

where $\phi(\cdot)$ and $\psi(\cdot)$ denote the fully convolutional encoder functions for HSI and LiDAR inputs, respectively. The outputs $\mathbf{F}^{\text{HSI}}$ and $\mathbf{F}^{\text{LiDAR}}$ represent the extracted nonlinear spatial features from the two modalities.

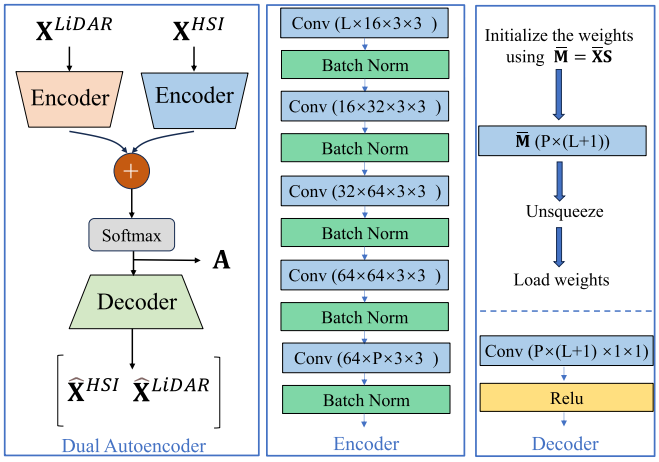

Fig. 3. Architecture of the autoencoder used for subpixel anchor graph learning for multimodel data.

These modality-specific features are fused via elementwise addition to form a unified latent representation

$$\mathbf{F}^{\text{Unified}} = \mathbf{F}^{\text{HSI}} + \mathbf{F}^{\text{LiDAR}}, \quad \mathbf{F}^{\text{Unified}} \in \mathbb{R}^{H\times W\times P}. \tag{22}$$

To obtain a subpixel anchor graph, we apply the Softmax function over the channel dimension

$$\mathbf{A} = \text{softmax}\left(\mathbf{F}^{\text{Unified}}\right). \tag{23}$$

This parameter-free operation efficiently enforces the abundance constraints of nonnegativity and sum-to-one across each pixel.

In the decoder stage, unlike conventional methods where decoder weights are randomly initialized, we leverage multimodal endmembers $\bar{\mathbf{M}}$ obtained in Sections III-B and III-C as initialization anchors. These anchors are loaded as learnable parameters in a $1 \times 1 \times L \times P$ convolutional kernel tensor $\bar{\mathcal{M}}$, which serves to reconstruct the original inputs from the estimated abundances

$$\bar{\mathbf{X}} = \mathcal{D}_{\bar{M}}(\mathbf{A}) \tag{24}$$

where $\mathcal{D}_{\bar{M}}$ denotes a decoder parameterized by $\bar{\mathcal{M}}$.

The forward computation of this network is equivalent to the formulation in (6). Notably, DSAGC is not entirely decoupled from the original SAGC; the optimal anchors derived in SAGC can be adopted to initialize the decoder endmembers and further refined via the deep framework, thereby incorporating spatial priors.

The entire network is trained end-to-end by minimizing a reconstruction loss over encoder parameters and the endmember tensor $\bar{\mathcal{M}}$, formulated as

$$\mathcal{L}\left(\phi, \psi, \bar{\mathcal{M}}\right) = \left\|\bar{\mathbf{X}} - \mathcal{D}_{\bar{M}}\left(\text{softmax}\left(\mathbf{F}^{\text{HSI}} + \mathbf{F}^{\text{LiDAR}}\right)\right)\right\|^2. \tag{25}$$

Finally, after obtaining the abundance maps via (23), we perform clustering following the same procedure described in Section III-E, consistent with the original SAGC pipeline.

V. Experiments

In this section, we evaluate the effectiveness of the proposed method on three real-world HS–LiDAR multisource

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TABLE II
LAND-COVER TYPES OF THE TRENTO, MUUFL GULFPORT, AND HOUSTON DATASETS

Trento		MUUFL Gulfport		Houston 2013	
Class Name	#Samples	Class Name	#Samples	Class Name	#Samples
Apple trees	4034	Trees	23246	Healthy grass	1251
Buildings	2903	Mostly grass	4270	Stressed grass	1254
Ground	479	Mixed ground surface	6882	Synthetic grass	697
Wood	9123	Dirt and sand	1826	Trees	1244
Vineyard	10501	Road	6687	Soil	1242
Rolds	3174	Water	2262	Water	325
		Building shadow	2233	Residential	1268
		Building	6240	Commercial	1244
		Sidewalk	1385	Road	1244
		Yellow curb	183	Highway	1227
		Cloth Panels	269	Railway	1233
				Parking Lot 1	1233
				Parking Lot 2	426
				Tennis Court	428
				Running Track	660
Total	30214	Total	53687	Total	15029

RS datasets. First, we compare our approach with state-of-the-art (SOTA) and efficient multimodal clustering algorithms. We also conduct parameter sensitivity analyses and ablation studies to further validate the method's effectiveness and robustness.

A. Datasets

The experimental datasets used in this section are as follows.

1) *Trento:* The Trento dataset includes HS and LiDAR data. The HS imagery was acquired using the AISA Eagle sensor, covering wavelengths from 0.40 to 0.98 μm with 63 spectral bands. The LiDAR data were collected with the Optech ALTM 3100EA airborne laser scanner. The imagery covers a rural area south of Trento, Italy, with a size of 600 $\times$ 166 pixels and 1-m spatial resolution, and includes six land-cover classes.
2) *MUUFL Gulfport*[1]: Collected over the University of Southern Mississippi—Gulfport campus in Long Beach, MS, USA. The HS image has 325 $\times$ 220 pixels and was acquired using the CASI-1500 sensor with 72 spectral channels (64 effective bands used), covering wavelengths from 375 to 1050 nm at 10-nm spectral resolution and 0.54 $\times$ 1.0 m spatial resolution. Corresponding LiDAR data were acquired with the Gemini ALTM sensor and processed into a digital surface model (DSM). A total of 11 land-cover classes are used in this scene.
3) *Houston*[2]: Released by the IEEE GRSS Data Fusion Contest (DFC), this dataset contains HS imagery with 349 $\times$ 1905 pixels and corresponding DSM data derived from LiDAR, covering the University of Houston and surrounding urban areas in TX, USA. The HS data have 144 spectral bands spanning 0.38–1.05 μm with a spatial resolution of 2.5 m.

The specific land-cover classes for the datasets are shown in Table II and Fig. 4. All modalities have been prealigned in spatial coordinates and maintain consistent spatial resolution across modalities to eliminate confounding factors in the experiments.

[1]https://github.com/GatorSense/MUUFLGulfport

[2]http://www.grss-ieee.org/community/technical-committees/datafusion/2013-ieee-grss-data-fusion-contest/

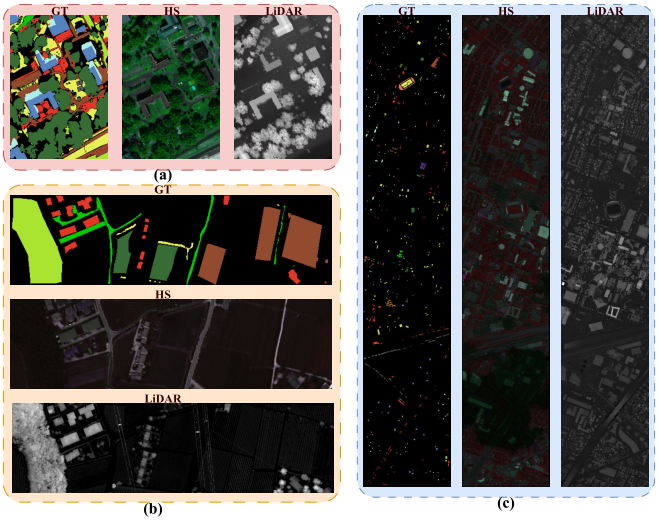

Fig. 4. Visualization of different multimodal RS datasets. Each group includes an HS pseudo-color image, a DSM derived from LiDAR, and the GT land-cover labels. (a) MUUFL Gulfport. (b) Trento. (c) Houston.

B. Baselines

In this section, we briefly introduce the SOTA multiview or multimodal clustering algorithms used for comparison with our proposed method.

1) *CCA [45]:* A classical statistical method for extracting shared information from two modalities. We use it to map HS–LiDAR data into a common space, followed by K-means clustering.
2) *MVKM [46]:* A classical multiview clustering baseline that performs K-means jointly across modalities under a unified framework.
3) *LMVSC [47]:* A representative multiview SC method that models on an anchor graph instead of a fully connected graph, achieving linear-time scalability.
4) *FPFC [48]:* A fuzzy clustering method designed for large-scale multimodal RS data, which leverages superpixel-level anchors to jointly learn a shared anchor graph and membership matrix.
5) *FPMVS-CAG [49]:* The first fast multiview SC algorithm with a balanced anchor strategy, featuring linear time complexity with respect to the number of samples.
6) *SMVSC [50]:* A scalable multiview SC method with unified anchors that integrates anchor learning and graph construction into a single optimization framework.
7) *MSGL [51]:* Based on anchor and bipartite graph ideas, it performs multiview clustering via structured graph learning.
8) *CDD [52]:* A recent diffusion-model-based multimodal generative clustering method that performs conditional control across modalities to derive joint clustering labels over the fused manifold.
9) *AMKSC [25]:* A recently proposed advanced multimodal anchor clustering method for RS data that applies spatial smoothing and kernel-space anchor learning to achieve SOTA performance with linear complexity.

C. Evaluating Metrics

In this article, we adopt five widely used evaluation metrics for clustering performance [53], including overall accuracy

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TABLE III
MULTIMODAL CLUSTERING PERFORMANCE OF THE SAGC COMPARED WITH DIFFERENT COMPARISON METHODS ON TRENTO DATASET

Class and Number	CCA	MVKM	LMVSC	FPFC	FPMVS-CAG	SMVSC	MSGL	CDD	AMKSC	SAGC	DSAGC
Apple trees #4034	64.4±2.6	28.2±2.5	44.7±6.0	0.0±0.0	23.8±7.0	47.3±8.8	0.0±0.0	60.4±2.7	**95.0±4.0**	83.2±0.4	89.8±5.8
Buildings #2903	82.0±7.1	0.0±0.0	77.5±0.2	28.1±7.3	0.0±0.0	0.0±0.0	84.5±4.8	15.7±7.1	88.7±2.4	90.3±1.2	**96.4±2.0**
Ground #479	0.0±0.0	15.9±3.9	0.0±0.0	**97.0±2.2**	7.7±6.7	12.9±3.9	0.0±0.0	11.4±2.2	0.0±0.0	73.0±4.6	0.0±0.0
Wood #9123	76.3±0.9	66.3±1.4	92.1±4.4	78.0±5.2	98.4±1.1	98.0±1.0	84.0±2.1	97.3±1.6	99.1±1.0	95.6±0.6	**100.0±0.0**
Vineyard #10501	48.9±3.1	50.8±2.1	55.2±7.0	89.0±3.3	73.0±8.4	65.5±0.7	**99.2±0.7**	96.0±2.5	93.1±2.3	66.9±0.9	97.7±0.6
Rolds #3174	63.6±2.1	**94.4±4.8**	58.7±9.1	71.0±2.2	90.5±7.2	90.6±8.4	24.7±1.9	75.4±6.1	84.0±4.4	78.3±1.9	91.5±3.9
ACC	63.2±2.6	51.6±3.7	66.6±4.1	66.2±3.2	67.9±2.8	72.9±2.9	70.5±1.9	80.4±1.6	92.6±0.9	81.3±0.2	**94.3±1.1**
Kappa	53.0±2.2	39.0±2.3	61.0±2.9	54.8±2.7	57.1±2.2	68.0±3.0	66.5±2.8	73.3±2.4	90.2±1.2	76.1±1.2	**92.4±1.4**
NMI	54.1±0.9	47.0±3.4	47.9±3.6	64.8±4.2	53.2±0.8	60.0±3.1	53.5±3.0	69.0±4.6	84.0±1.4	69.6±3.3	**86.4±3.0**
ARI	44.0±1.1	34.7±2.9	38.7±3.7	57.9±3.3	53.3±2.6	59.6±3.1	52.2±1.6	72.7±5.7	87.9±1.3	67.9±1.2	**91.2±1.9**
Purity	72.3±4.7	67.6±3.5	68.9±5.0	71.9±3.1	72.0±2.4	74.7±3.3	71.7±2.9	80.7±1.5	93.2±0.6	82.9±0.2	**94.3±1.0**
Time (s)	0.294	1.969	5.1673	8.4985	212.8796	200.5626	234.1521	710.34	83.545	81.22	88.48

(ACC), Kappa coefficient (Kappa), normalized mutual information (NMI), adjusted Rand index (ARI), and cluster purity (Purity). Among them, ACC, NMI, and Purity are bounded within $[0, 1]$, while Kappa and ARI range from $[-1, 1]$. A higher score across all metrics indicates better clustering performance. Moreover, since clustering is an unsupervised task, we employ the Hungarian algorithm [54] to assign the optimal matching between predicted and ground-truth (GT) labels, ensuring the consistency and reliability of the analysis.

D. Implementation Details

We implemented the proposed SAGC and its extended variant DSAGC in Python. The experiments were conducted on a platform that includes an NVIDIA RTX 2080Ti GPU with 11 GB of memory and an Intel[3] Xeon[3] Gold 5218 CPU at 2.30 GHz with 128-GB RAM, running Linux. All hyperparameters were selected via grid search to obtain optimal initialization, where the search process was guided by (6). Specifically, we explored the hyperparameter K that determines the anchor/endmember library size (see Algorithm 1) over the range $[1, 2, \ldots, 13]$, the regularization parameter λ controlling L_1 strength over $[10^{-1}, 10^{-2}, \ldots, 10^{3}, 10^{4}]$, and the sparsity-enforcing parameter α also over $[10^{-1}, 10^{-2}, \ldots, 10^{3}, 10^{4}]$. In the DSAGC method, the Adam optimizer was utilized to update network parameters, and the optimal order of learning rate was determined via grid search over the range $[10^{-1}, 10^{-2}, \ldots, 10^{-10}]$. Due to the varying complexity of land-cover distributions across different datasets, the learning rates for the three datasets were set to 0.1, 0.1, and 0.2, respectively, based on the search results. The maximum number of iterations for both methods was set to 200. For fairness and reproducibility, all experiments were repeated ten times with different random seeds, and the reported results correspond to the mean and standard deviation. All competing methods were run using the optimal settings provided in their respective articles.

E. Quantitative Results

Tables III–V present the quantitative results on the three datasets, including the clustering ACC for each class, the average values of five overall clustering metrics, and the total runtime of each algorithm. For readability, the best results are highlighted in bold, and the second-best results are marked with underline.

Table III shows the quantitative comparison results on the Trento dataset. Using ACC as a representative metric, the proposed SAGC demonstrates clear competitiveness and acceptable runtime compared to all baselines except AMKSC. The relatively weaker performance of SAGC against AMKSC on this dataset is primarily due to AMKSC's use of the spatial smoothing and kernel method, resulting in significantly higher accuracy (92.6%) compared to other methods. This highlights the importance of modeling nonlinear and spatial features. However, this design introduces substantial optimization complexity, leading to increased time costs when the sample size is large. It is also important to note that this specialized design does not guarantee consistent improvements across datasets. On the MUUFL and Houston datasets, AMKSC lags behind SAGC by 12.8% and 7.5%, respectively. By using general-purpose deep learning optimizers like Adam, DSAGC avoids complex optimization procedures and demonstrates stronger generalization, achieving top-tier accuracy and runtime performance across most metrics on the three datasets.

Moreover, compared to general anchor-graph-based SC baselines (such as LMVSC and SMVSC), SAGC consistently achieves superior results. The reason is that LMVSC and SMVSC are designed for general multisource data, without considering the subspace similarity issues caused by spectral mixing effects. In contrast, our method learns subpixel abundances as anchor graphs, explicitly modeling the subpixel information from mixed spectra to endmembers. This leads to significant improvements across all three datasets (with an average accuracy increase of about 10% compared to LMVSC and SMVSC).

Notably, the quantitative results in Tables III–V show a substantial number of 0.0%, indicating the presence of empty clusters in certain classes (i.e., all samples of a class are completely misclassified). Empty clusters are generally caused by class imbalance, where dominant classes govern the clustering process, causing extremely small classes to be entirely misclassified. In addition, high similarity in spectral or LiDAR features can reduce interclass separability, leading to the misassignment of land-cover types with similar characteristics. Although MSGL mitigates subspace

[3]Registered trademark.

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TABLE IV

MULTIMODAL CLUSTERING PERFORMANCE OF THE SAGC COMPARED WITH DIFFERENT COMPARISON METHODS ON MUUFL GULFPORT DATASET

Class and Number	CCA	MVKM	LMVSC	FPFC	FPMVS-CAG	SMVSC	MSGL	CDD	AMKSC	SAGC	DSAGC
Trees #23246	21.6±3.6	29.9±3.6	45.4±2.6	32.8±2.2	47.6±3.2	45.8±5.4	**81.4±2.1**	28.2±7.1	38.1±1.7	44.7±1.1	80.0±6.2
Mostly grass #4270	51.1±7.5	1.3±0.0	**85.2±4.8**	42.2±3.2	3.4±0.2	79.6±1.3	25.6±1.1	10.5±8.2	65.3±0.4	80.4±2.1	33.0±3.9
Mixed ground surface #6882	58.7±8.2	64.9±6.1	48.2±5.8	51.9±4,2	7.0±4.8	20.7±0.9	8.4±5.2	**76.0±6.3**	12.7±2.1	51.4±0.9	66.3±4.1
Dirt and sand #1826	2.9±0.9	48.2±3.2	59.5±5.0	38.0±1.6	0.8±0.0	1.4±1.2	**81.4±0.7**	0.9±0.0	61.5±4.9	28.9±9.2	50.3±3.0
Road #6687	46.2±8.1	58.4±1.4	**94.7±6.7**	43.9±2.2	58.2±4.8	57.1±2.5	0.0±0.0	85.4±2.1	70.6±7.0	76.3±3.7	77.8±1.8
Water #2262	**91.2±7.3**	0.0±0.0	0.9±0.0	20.1±8.9	0.2±0.0	0.0±0.0	7.0±0.8	1.5±0.5	7.7±3.2	71.6±1.2	6.3±1.1
Building shadow #2233	37.0±6.9	15.8±7.1	76.6±4.3	14.2±7.9	88.9±1.6	0.4±0.0	4.2±0.8	67.5±5.5	44.7±5.6	**90.9±1.7**	23.7±3.7
Building #6240	40.2±4.5	7.0±4.2	32.8±3.2	42.1±0.7	12.4±0.6	13.6±1.8	0.0±0.0	50.5±2.1	62.5±4.9	**71.2±2.7**	66.8±1.5
Sidewalk #1385	0.5±0.0	4.5±0.4	1.2±0.0	3.7±1.1	1.6±0.0	0.2±0.0	0.0±0.0	**74.6±1.5**	8.6±3.6	45.5±10.2	4.1±3.3
Yellow curb #1385	0.0±0.0	0.0±0.0	1.1±0.3	0.0±0.0	0.5±0.0	3.3±1.4	0.0±0.0	0.0±0.0	**9.8±7.2**	1.8±0.3	0.1±0.1
Cloth panels #269	0.0±0.0	0.0±0.0	**92.2±3.4**	0.0±0.0	0.0±0.0	0.3±0.0	0.0±0.0	26.0±10.4	0.0±0.0	62.0±8.2	0.0±0.0
ACC	33.8±3.6	31.9±3.3	54.0±4.6	36.8±5.3	34.2±1.6	39.2±2.4	58.7±0.9	44.1±5.2	43.6±4.1	56.4±0.8	**66.0±3.4**
Kappa	25.0±2.4	18.4±4.2	49.6±2.5	25.6±3.3	18.8±3.5	34.3±2.1	**55.7±1.8**	36.8±4.1	35.4±1.7	49.1±1.2	55.4±3.4
NMI	30.0±2.8	18.9±2.5	49.6±4.2	23.3±4.6	27.7±1.1	37.7±3.0	37.3±1.5	50.1±2.2	40.0±3.0	**50.4±2.1**	49.8±1.0
ARI	13.3±4.9	7.8±5.2	41.6±1.3	14.5±3.4	19.6±3.1	25.7±6.2	40.2±3.2	30.3±3.4	27.1±1.5	32.3±3.8	**50.4±4.6**
Purity	57.2±3.9	46.5±2.7	**75.5±2.2**	54.3±4.0	51.5±1.9	59.5±2.9	63.5±1.9	63.7±4.2	66.8±4.6	70.6±0.7	72.0±1.7
Time (s)	0.401	2.214	7.9824	7.4392	508.0643	568.8984	300.6835	314.78	72.799	68.286	79.454

TABLE V

MULTIMODAL CLUSTERING PERFORMANCE OF THE SAGC COMPARED WITH DIFFERENT COMPARISON METHODS ON HOUSTON DATASET

Class and Number	CCA	MVKM	LMVSC	FPFC	FPMVS-CAG	SMVSC	MSGL	CDD	AMKSC	SAGC	DSAGC
Healthy grass #1251	20.6±6.1	85.8±8.1	42.5±3.7	85.0±6.4	0.0±0.0	0.0±0.0	54.3±3.1	81.2±1.2	92.5±5.7	79.8±0.4	**95.4±6.1**
Stressed grass #1254	8.2±0.1	0.0±0.0	20.9±2.1	53.7±7.9	0.0±0.0	1.1±0.0	**70.0±4.1**	14.2±2.1	10.4±2.2	41.6±4.4	9.5±3.1
Synthetic grass #697	18.8±9.2	0.0±0.0	24.5±3.9	0.0±0.0	0.0±0.0	0.0±0.0	98.7±1.0	93.1±3.5	89.4±7.2	70.0±3.2	**100.0±0.0**
Trees #1244	26.1±4.6	72.9±4.9	41.1±0.3	83.2±2.1	89.8±10.4	89.4±5.7	50.9±2.9	65.3±0.7	91.5±10.2	75.2±2.1	**96.3±3.1**
Soil #1242	0.0±0.0	95.0±2.7	43.2±1.7	**100.0.±0.0**	86.7±1.3	70.9±4.5	61.0±3.4	97.1±5.2	19.1±5.3	76.0±2.4	88.1±9.1
water #325	0.0±0.0	0.0±0.0	83.7±3.2	7.3±4.1	0.0±0.0	0.0±0.0	0.0±0.0	73.2±1.2	8.0±0.9	**85.5±0.0**	44.0±6.9
Residential #1268	16.9±6.2	78.3±6.4	42.0±3.3	3.8±3.0	2.0±1.2	**91.1±0.3**	59.8±2.1	65.0±2.7	27.0±4.5	54.1±3.8	75.5±5.6
Commercial #1244	21.2±2.4	22.1±2.7	47.8±7.0	14.4±9.1	0.9±0.0	5.5±0.9	36.0±8.2	20.3±0.5	45.3±5.2	17.2±1.0	**49.2±8.2**
Road #1244	30.0±0.0	0.0±0.0	0.0±0.0	**88.5±5.7**	4.7±1.7	21.1±2.7	67.9±2.5	78.6±5.6	67.0±3.6	68.6±4.1	7.8±0.2
Highway #1227	8.0±1.5	26.6±3.5	**46.3±4.3**	26.6±1.2	0.6±0.0	0.5±0.0	27.3±5.9	0.0±0.0	28.0±3.1	16.8±5.1	17.8±5.1
Railway #1233	70.0±7.8	18.0±5.1	11.6±4.2	0.0±0.0	**89.9±6.2**	10.4±5.2	0.0±0.0	3.1±1.0	64.0±8.0	44.3±0.4	63.0±7.8
Parking Lot 1 #1233	23.7±2.5	26.9±7.4	32.1±0.5	9.1±2.9	3.7±1.4	7.1±0.9	0.0±0.0	1.3±0.5	32.9±2.1	**33.7±1.0**	0.0±0.0
Parking Lot 2 #425	0.0±0.0	0.0±0.0	3.4±0.0	0.0±0.0	0.0±0.0	3.0±0.5	1.9±0.2	0.0±0.0	0.0±0.0	0.0±0.0	**47.8±3.3**
Tennis Court #428	0.0±0.0	21.2±2.3	0.0±0.0	0.0±0.0	73.1±5.1	67.1±1.7	59.6±2.4	23.7±4.9	0.0±0.0	58.6±3.2	**80.6±4.5**
Running Track #660	99.1±0.9	0.0±0.0	91.8±5.6	**99.8±0.0**	80.0±6.3	94.4±0.5	0.0±0.0	0.0±0.0	0.0±0.0	55.9±5.2	65.2±9.2
ACC	23.8±5.0	35.9±5.0	34.2±2.5	43.0±1.9	28.6±1.4	30.9±3.9	41.9±2.7	45.2±2.2	43.9±3.1	51.4±0.8	**56.0±3.2**
Kappa	18.1±3.3	30.1±4.0	29.1±2.1	38.4±1.6	22.5±3.4	25.0±3.6	37.2±4.5	41.6±4.1	39.2±4.1	47.5±2.2	**52.8±3.1**
NMI	32.0±3.2	44.8±3.1	41.7±1.9	59.5±4.6	39.9±2.3	44.8±4.5	52.8±3.3	55.2±3.7	49.2±1.1	51.9±1.1	**63.2±4.7**
ARI	10.3±2.9	23.0±3.0	17.0±4.8	25.7±4.2	17.6±1.5	19.7±3.4	27.6±4.1	35.2±2.6	30.2±2.7	32.7±0.2	**42.3±2.1**
Purity	27.8±3.4	36.3±3.3	40.8±3.1	48.9±3.2	30.3±4.5	34.6±3.5	48.4±2.1	47.1±1.5	46.5±2.4	52.1±0.5	**56.6±2.9**
Time (s)	0.977	53.512	140.2415	55.9735	766.426	694.8678	4716.541	974.252	629.595	302.909	333.41

similarity caused by spectral mixing by explicitly constraining the number of subspaces and achieves competitive performance on the MUUFL dataset (ACC of 58.7%), this approach of enforcing subspace counts through rank constraints can result in the complete disappearance of certain class subspaces. Consequently, MSGL experiences the most clustering collapses (empty clusters) on the MUUFL dataset, mainly in categories such as Road, Building, Sidewalk, Yellow curb, and Cloth panels, with similar trends observed in other datasets. Although MSGL achieves a slight ACC improvement (about 2.3% higher than SAGC), its practical value is limited by the large number of empty clusters it produces. In contrast, SAGC and DSAGC fundamentally avoid the negative impact of spectral mixing by directly modeling subpixel anchor graphs from mixed pixels to pure endmembers, better preserve modeling capacity, and maintain competitive accuracy across all datasets with fewer empty clusters, demonstrating strong practical potential. In addition, on the more complex Houston dataset, DSAGC achieves overall superior performance, while SAGC attains the second-best results across multiple metrics. Although its NMI score is lower than that of FPFC, this is primarily due to different sensitivities of NMI and ACC to class imbalance. Considering all metrics comprehensively, both SAGC and DSAGC demonstrate clear advantages even in complex scenarios.

Overall, although earlier algorithms achieve better runtime efficiency, their clustering accuracy remains unsatisfactory. Compared with more recent baselines, SAGC and DSAGC exhibit varying degrees of strengths and weaknesses in per-class accuracy, which can be attributed to the complexity

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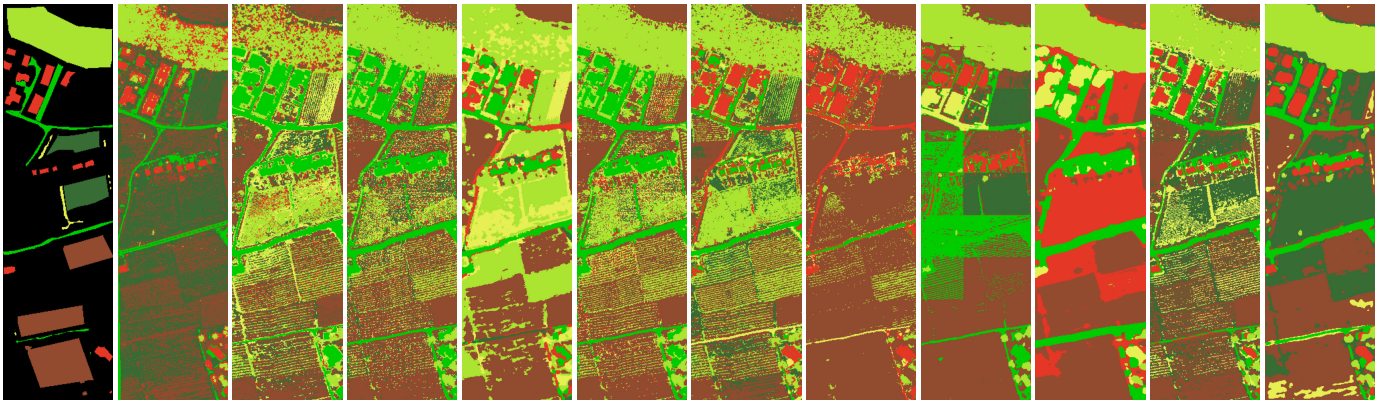

Fig. 5. Clustering results of Trento Gulfport dataset. (a) GT. (b) CCA. (c) MVKM. (d) SMVSC. (e) FPFC. (f) FPMVS-CAG. (g) LMVSC. (h) MSGL. (i) CDD. (j) AMKSC. (k) SAGC. (l) DSAGC. (See also Table III.)

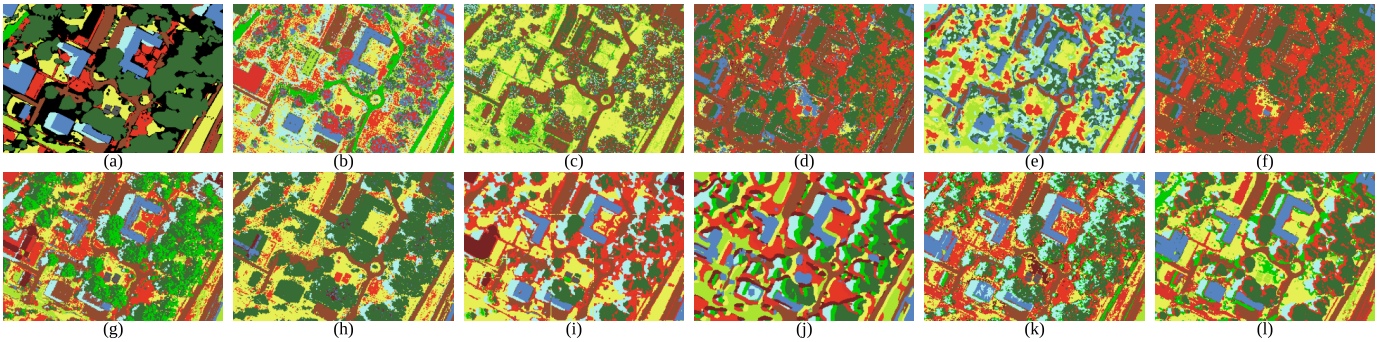

Fig. 6. Clustering results of MUUFL Gulfport dataset. (a) GT. (b) CCA. (c) MVKM. (d) SMVSC. (e) FPFC. (f) FPMVS-CAG. (g) LMVSC. (h) MSGL. (i) CDD. (j) AMKSC. (k) SAGC. (l) DSAGC. (See also Table IV.)

of land-cover distributions across datasets and the inherent model biases favoring certain categories. Nevertheless, in terms of both overall evaluation metrics and runtime, our proposed methods consistently achieve superior accuracy and computational efficiency across all three datasets. This demonstrates strong generalization ability and robustness, and further underscores the practical importance of explicitly addressing spectral mixing in HS–LiDAR clustering.

F. Qualitative Results

Because RS data exhibit strong spatial homogeneity, qualitative visual inspection complements quantitative comparisons by highlighting differences between methods. Figs. 5–7 show the cluster labeling results projected back into the original image space. Overall, all baseline methods capture some spatial distribution consistent with land-cover classes, but differ in detail. For example, on the Trento dataset, methods such as CDD, AMKSC, and DSAGC show excellent spatial continuity, largely because they explicitly model spatial information. In contrast, many baselines produce significant salt-and-pepper noise, reflecting their inability to preserve RS-specific spatial structure.

Interestingly, among methods that do not explicitly model spatial information, our proposed SAGC achieves the best spatial continuity. As shown in Fig. 5, for the Wood class, SAGC produces overall continuous results comparable to those of spatially aware methods. This supports our theoretical analysis (see Section III-C), suggesting that jointly learning HS–LiDAR endmembers and a shared abundance representation serves as an implicit spatial regularization mechanism (discussed further in Section V-G). While this highlights the benefit of multimodal abundance learning, regions with weak LiDAR spatial cues (e.g., Vineyard) still show discontinuities, underscoring the need for explicit spatial modeling.

In addition, in RS imagery, clustering building boundaries is particularly challenging due to spectral mixing at building edges. The quantitative results in Tables III and IV show that SAGC and DSAGC significantly outperform all baseline methods for the building class (e.g., on the MUUFL dataset, CDD achieves only 50.5% ACC, while SAGC achieves 71.2% and DSAGC achieves 66.8%). By focusing on the building cluster maps in Figs. 5 and 6, our method shows strong alignment with the GT, particularly at the boundaries. In contrast, AMKSC's over-smoothing causes building edge samples to be overwhelmed by surrounding pixels. By leveraging a lightweight fully convolutional encoder for spatial feature extraction, DSAGC achieves more stable and accurate clustering results. In addition, although our method achieves segmentation results on the Trento dataset that are very close to the GT, there remains a noticeable gap in more complex scenarios such as Houston. As shown in Fig. 7, the main reason is the severe imbalance between the target classes to be clustered and the total number of input samples in such complex scenes. Under an unsupervised setting, this inherent intraclass imbalance cannot be corrected or compensated for

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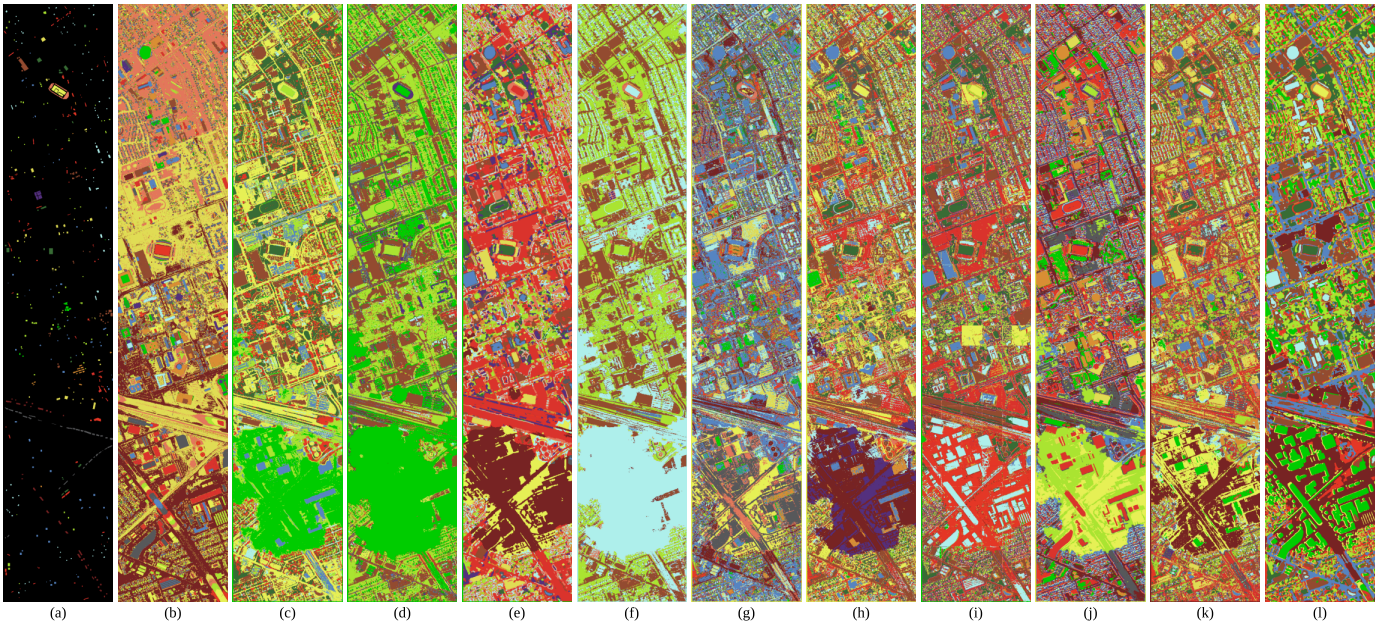

Fig. 7. Clustering results of Houston Gulfport dataset. (a) GT. (b) CCA. (c) MVKM. (d) SMVSC. (e) FPFC. (f) FPMVS-CAG. (g) LMVSC. (h) MSGL. (i) CDD. (j) AMKSC. (k) SAGC. (l) DSAGC. (See also Table V.)

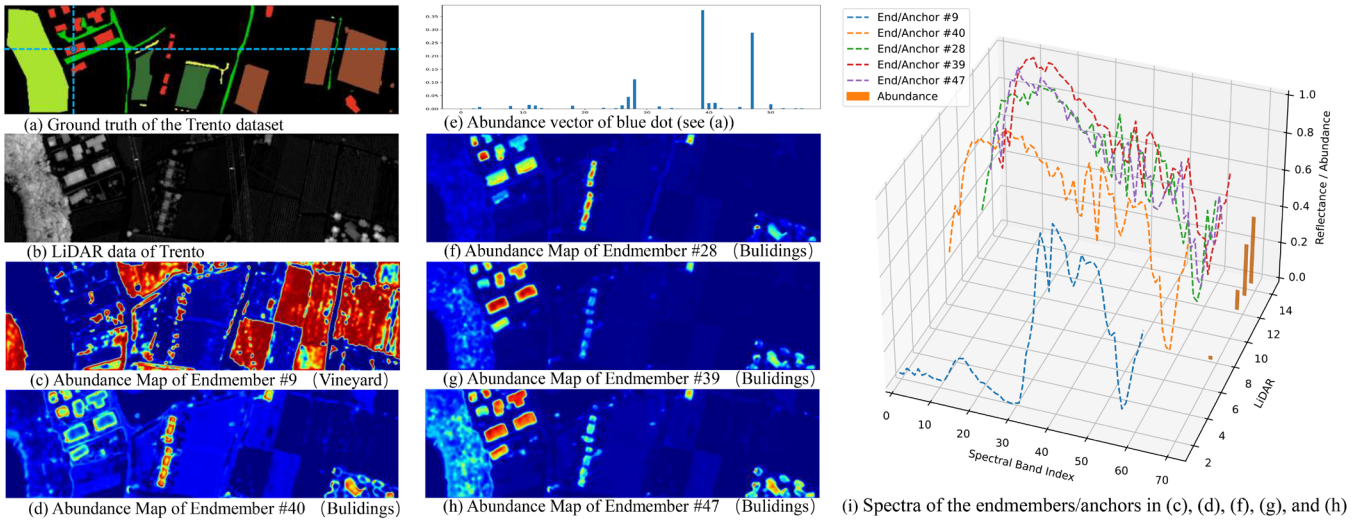

Fig. 8. Visualization of the relationship between samples in the Trento dataset and a subset of endmembers/anchors. (a) GT. (b) LiDAR data showing spatial priors. (c) Abundance of the vineyard endmember. (d) and (f)–(h) Abundance of endmembers corresponding to different buildings. (e) Endmember spectra of the blue-dotted samples in (a). (i) Spectra of the endmembers/anchors in (c), (d), and (f)–(h), along with their corresponding LiDAR elevations. Orange bars indicate the abundance of these endmembers/anchors for the blue-dotted samples in (a).

using label distributions, which remains a major challenge in clustering tasks.

G. Visualization of the Unified Subpixel Anchor Graph

To verify the assertion in Section III-C that, under multimodal endmember conditions, a pixel's spectrum is primarily a mixture of endmembers with similar LiDAR elevation values, we design the following experiment. Using the proposed SAGC method on the Trento dataset, we first learn the subpixel anchor graph and then select the spectrum of a building pixel [marked as a blue dot in Fig. 8(a)] as the target. We visualize its abundance vector in Fig. 8(e), confirming its sparsity, which enhances clustering discriminability. Next, we select several activated endmembers (#9, #28, #39, #40, and #47), and show their spectral curves, LiDAR elevation values, and abundance weights in Fig. 8(i). We observe that endmembers #28, #39, and #47, which have high abundances, have similar LiDAR values [as seen in their positions in Fig. 8(i)], while #40, with a different LiDAR value but similar spectrum, has an almost zero abundance. This verifies the assertion that only LiDAR-similar endmember subsets are activated.

In addition, Fig. 8(c), (d), and (f)–(h) show the spatial distribution of different endmembers' contributions across all samples, providing qualitative insights into multimodal

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TABLE VI
ABLATION STUDY OF THREE CORE CONSTRAINTS IN SUBPIXEL ANCHOR LEARNING ON HOUSTON DATASET (%)

$\mathbf{A}_{pn} \geq 0$	$\mathbf{a}_n^T\mathbf{1} = 1$	$\mathbf{S}_{np} \geq 0$	ACC	Kappa	NMI	ARI	Purity
			30.7	25.7	45.1	14.4	33.0
✓			35.7	31.1	40.4	20.2	37.8
	✓		29.0	23.0	31.3	18.4	33.3
		✓	41.9	37.1	45.7	26.7	43.8
✓	✓		47.3	43.5	**59.0**	**33.4**	49.6
✓		✓	44.4	39.8	46.1	29.9	45.6
	✓	✓	40.0	35.3	35.3	26.7	41.1
✓	✓	✓	**51.4**	**47.5**	51.9	32.7	**52.1**

subpixel anchor graph learning. Specifically, Fig. 8(i) shows that endmembers #28, #39, #40, and #47 have highly similar spectral characteristics, representing building materials. However, different LiDAR elevation values for different buildings result in distinct abundance contribution patterns. These results suggest that multimodal subpixel anchor graph learning can partition the same land-cover class into multiple anchors at the subpixel level, enhancing clustering robustness for samples within the same class. Furthermore, Fig. 8(d) qualitatively shows how SAGC can more accurately identify building edge samples by utilizing the subpixel unmixing mechanism. Fig. 8(c) shows that when the spectral curves of endmembers are completely different, their contribution distributions also differ entirely, indicating that even with an overcomplete set of anchors, SAGC is still capable of learning representative anchors.

H. Parameter Sensitivity and Ablation Studies

In this section, we systematically analyze the impact of different hyperparameters on the performance of our method and conduct ablation studies on its key components to verify their effectiveness.

1) Ablation Study on Core Constraints: To validate the necessity of the three physically meaningful subpixel information constraints, we conducted ablation experiments on the challenging Houston dataset. An unconstrained model was used as the baseline to assess the individual impact of each constraint on model performance. As shown in Table VI, introducing the abundance nonnegativity constraint alone leads to a notable performance gain (+5.0% ACC). While applying the sum-to-one constraint alone yields limited improvement in clustering accuracy, its combination with the nonnegativity constraint results in a substantial enhancement (+16.6% ACC). Additionally, note that the constraint $\mathbf{S}_{\text{np}} \geq 0$ refers to the introduction of the anchor selection matrix $\mathbf{S}$ as a dynamically updated parameter, constrained to preserve its physical interpretation as endmember candidates. This setting outperforms the baseline significantly (+11.2% ACC) and consistently improves performance when combined with any abundance constraint, demonstrating its necessity. The optimal result is achieved when all three constraints are jointly applied, verifying that each component is indispensable for effective subpixel anchor learning.

2) Ablation Study on Anchor Selection: To evaluate the effectiveness of the proposed MADSS, we compare the clustering performance of different anchor selection methods within the proposed framework. Note that, to comprehensively assess

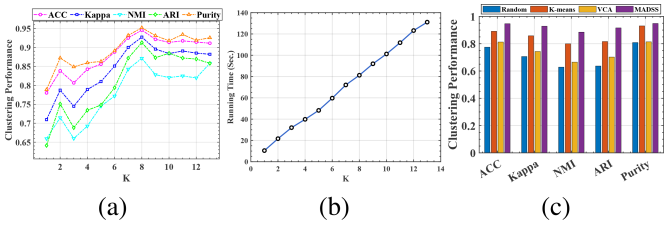

Fig. 9. Impact of anchor initialization on clustering performance and runtime. (a) and (b) Use our MADSS strategy. (a) Impact of different over-complete anchor ratios K (see Section III-B) on clustering performance. (b) Influence of anchor number. (c) Comparison of random, K-means, and VCA initialization.

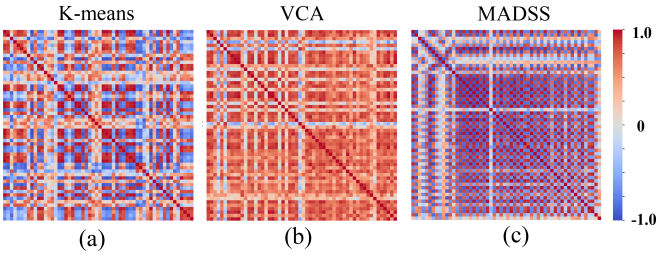

Fig. 10. Comparison of anchor redundancy across different selection methods. (a) K-means. (b) VCA. (c) MADSS.

the impact of anchor strategies on nonlinear spectral representations, spatial structures, and runtime, we report the results based on the DSAGC. As shown in Fig. 9(c), the random initialization yields the worst performance, as our framework heavily relies on the quality of the initial subpixel anchors. Randomly selected candidates fail to represent physically meaningful endmembers, leading to suboptimal clustering. The performance improves slightly when using VCA, but remains inferior compared to directly applying K-means. In contrast, our proposed method adopts a diversity-maximizing selection strategy that selects anchors from the combined candidate pool generated by VCA and K-means. This approach achieves the best results across all five evaluation metrics, demonstrating the effectiveness of our strategy.

3) Anchor Redundancy Analysis: To further support the assertion in Section III-B, we analyze the redundancy of anchors selected by VCA, K-means, and MADSS by computing pairwise cosine distances. As shown in Fig. 10, the redundancy matrix is presented, VCA selects many similar (highly redundant) endmembers under the overcomplete setting, lacking intercluster diversity. K-means provides lower redundancy but fails to explicitly capture subpixel representations. In contrast, MADSS greedily selects anchors from the combined VCA and K-means candidates, resulting in significantly lower redundancy while preserving both endmember representativeness and anchor diversity, thus ensuring superior clustering performance.

4) Impact of the Number of Anchors: In Algorithm 1, we set the number of anchors equal to the number of endmembers estimated by K-Hysime. Fig. 9(a) and (b) shows the impact of different K values on clustering performance and runtime. Smaller K values result in poorer performance, likely due to insufficient coverage of representative subpixel endmembers. As K increases, performance improves and peaks at $K = 8$ on the Trento dataset, then stabilizes. This indicates that once K exceeds a certain threshold, the K-times hysime-based

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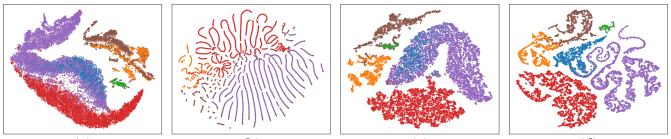

Fig. 11. T-SNE visualization of (a) HS data, (b) LiDAR data, and (c) learned abundance. (d) Learned abundance by deep learning.

TABLE VII

RESULTS (%) OF DIFFERENT CLUSTER WAYS ON THREE DATASETS. OOM DENOTES OUT OF MEMORY

Datasets	Cluster Ways	ACC	Kappa	NMI	ARI	Purity
Trento	EVD and K-means	93.0	90.7	85.5	89.7	93.8
	SVD and K-means	95.4	93.6	88.2	92.3	96.2
	Our method	94.3	92.4	86.4	91.2	94.3
MUUFL	EVD and K-means	56.0	39.1	39.5	37.5	56.1
	SVD and K-means	58.4	46.1	43.2	37.9	61.8
	Our method	66.0	55.4	49.8	50.4	72.0
Houston	EVD and K-means	OOM	OOM	OOM	OOM	OOM
	SVD and K-means	51.9	48.4	60.5	39.7	55.0
	Our method	56.0	52.8	63.2	42.3	56.6

method maintains stable performance, reducing reliance on prior knowledge for anchor selection. Additionally, Fig. 9(b) shows that runtime grows approximately linearly with K, allowing the use of smaller K (e.g., $K = 2$ on Houston) to balance accuracy and efficiency, with a significantly lower cost than AKMSC.

5) Impact of Nonlinear and Spatial Features: In the Introduction, we assert that subpixel information obtained from unmixing can improve clustering performance, and in Section IV, we emphasize the importance of nonlinear and spatial feature modeling, which motivates the proposed deep learning extension. To validate these motivations and assertions, we use t-distributed stochastic neighbor embedding (t-SNE) to visualize the abundance vectors derived from HS data, LiDAR data, SAGC, and its deep learning extension DSAGC in 2-D space (see Fig. 11). The results indicate that the raw data exhibit significant class overlap due to subspace similarity caused by pixel mixing, while the abundance representations obtained by SAGC show improved interclass separability but still have some overlap (e.g., blue and purple samples). By incorporating nonlinear and spatial feature learning, DSAGC further eliminates interclass overlap, demonstrating the importance of such features in SAGC.

6) Impact of Cluster Ways: To assess the impact of different cluster ways, we report performance on three datasets in Table VII. The results show that our proposed approach, which directly applies AAD-based K-means in the column space of the anchor graph $\mathbf{A}$, achieves comparable or even superior performance to methods that perform EVD on the full graph or SVD on the anchor graph to obtain optimal embeddings before clustering. Notably, our method uses standard K-means with a time complexity of $\mathcal{O}(\text{TPND})$, where T is the number of iterations and D is the number of clusters, with $P << N$. Therefore, the method not only avoids complex SVD and spectral decomposition steps but also maintains scalability

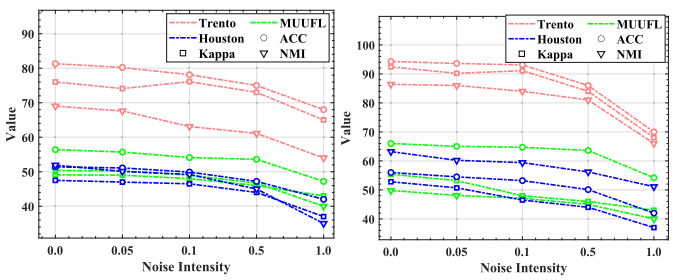

Fig. 12. Robustness of (a) SAGC and (b) DSAGC under varying noise intensities.

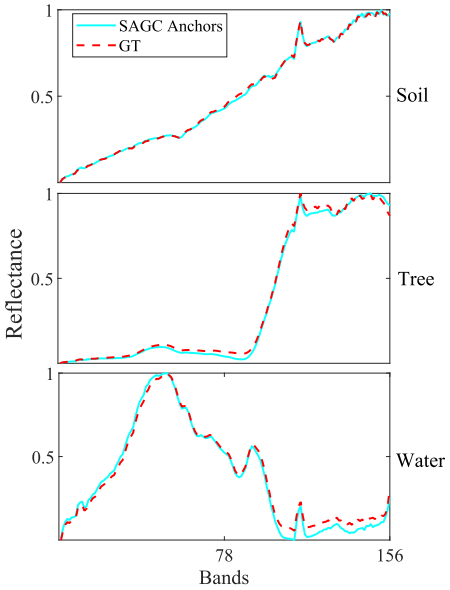

Fig. 13. Comparison between the optimal anchors obtained by applying the SAGC algorithm to the Samson dataset and the GT reference endmember spectra of different materials.

on large-scale datasets comparable to anchor graph-based approaches.

7) Analysis of Robustness to Noise: To evaluate the robustness of the proposed SAGC and DSAGC under noisy conditions, we plotted the clustering performance across different noise intensities for all three datasets (see Fig. 12). As shown in the figure, under low noise levels (noise intensity ≤ 0.1), the evaluation metrics of SAGC and DSAGC remain almost identical to those obtained on noise-free data, indicating that both methods maintain strong clustering performance under mild noise perturbations. As the noise intensity increases (noise intensity ≥ 0.5), the performance gradually decreases, but overall remains stable. These results demonstrate that SAGC and DSAGC are capable of handling various levels of noise interference and exhibit good robustness.

8) Physical Interpretability: To validate the advantage of our method in terms of the physical interpretability of the learned anchors, we designed a comparative experiment on anchor spectral curves. Since widely used clustering datasets do not explicitly consider spectral mixing and therefore lack GT subpixel reference spectra, we conducted a fidelity com-

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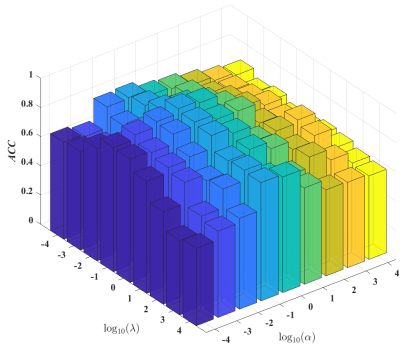

Fig. 14. Clustering accuracy under different values of $\log_{10}(\lambda)$ and $\log_{10}(\alpha)$.

parison experiment on the classical HS unmixing dataset Samson [31]. We applied the proposed method to the Samson dataset and compared the refined optimal anchors obtained after the update process with the provided reference pure spectra. Fig. 13 shows that the anchors successfully preserve the spectral curve shapes, demonstrating the strong spectral fidelity of our approach.

9) Sensitivity of λ and α: We conduct a sensitivity analysis on the logarithmic scales of parameters λ and α, denoted as $\log_{10}(\lambda)$ and $\log_{10}(\alpha)$, which, respectively, control the update strengths of the sum-to-one constraint and the sparsity regularization. Fig. 14 shows the clustering performance under varying logarithmic values of the two parameters. It can be observed that the clustering ACC remains stable across a broad range of values, gradually improving as $\log_{10}(\lambda)$ and $\log_{10}(\alpha)$ increase from negative to near-zero values. The best performance is achieved when $\log_{10}(\lambda) = -1$ and $\log_{10}(\alpha) = -1$, corresponding to $\lambda = 0.1$ and $\alpha = 0.1$. This suggests that moderate regularization strength in a logarithmic scale leads to optimal performance. Both overly small and overly large values can harm the results, indicating the necessity of careful balance.

VI. Conclusion

This article unifies HU and AGC to propose a novel SAGC algorithm for multimodal HS–LiDAR clustering. Our method enables end-to-end learning of subpixel anchor graphs, fundamentally mitigating spectral mixing effects. We design an anchor selection strategy, MADSS, to ensure anchor diversity and subpixel representativeness. By leveraging abundance information, we inject LiDAR spatial priors into HS data to achieve implicit spatial regularization. The proposed method scales linearly with the number of samples, ensuring good scalability. To efficiently model spatial and nonlinear features, we further introduce an extended version, DSAGC, which employs a lightweight dual convolutional autoencoder for more stable and generalized SAGC. Extensive experiments on three real HS–LiDAR datasets demonstrate superior clustering performance and runtime efficiency, and ablation and visualization studies validate the motivation and effectiveness of our approach. Future work will focus on improving the generalization of subpixel anchor graphs for out-of-sample clustering tasks. Our method also has some limitations, including the inability to perform out-of-sample predictions and the inability to address the impact of data imbalance on clustering performance. Future work will focus on improving the generalization of subpixel anchor graphs for out-of-sample clustering tasks and clustering on imbalanced data.

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Yang Liu received the B.S. degree from the School of Computer Science, Chengdu Normal University, Chengdu, China, in 2022. He is currently pursuing the M.E. degree in artificial intelligence with the School of Computer Science and Cyber Security, Chengdu University of Technology, Chengdu.

His research interests include hyperspectral image processing and computer vision.

Shujun Liu (Member, IEEE) received the Ph.D. degree from Chengdu University of Technology, Chengdu, China, in 2024.

From 2024 to 2025, he was a Visiting Ph.D. Student with ITC, University of Twente, Enschede, The Netherlands. Since 2025, he has been a Lecturer with the School of Computer Engineering, Chengdu Technological University, Chengdu. His research interests lie in machine learning and image processing.

Luxi Xiao received the B.S. degree in applied physics from the University of Science and Technology Liaoning, Anshan, China, in 2023. He is currently pursuing the M.E. degree in geophysics with Chengdu University of Technology, Chengdu, China.

His research interests include multimodal remote sensing and unsupervised clustering.

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